TEXORA · LECTURE SHEET

3D Woven and Knitted Fabrics

Fabrics with real thickness architecture: yarns travelling in three directions, layers bound through their depth, and near-net shapes off the machine — the textile answer to delamination in composites and to cut-and-sew waste in shaped products.

FM-4103 · SPECIAL FABRIC PRODUCTION · TOPIC 5 · FABRIC MANUFACTURING SPECIALIZATION

1. From 2D cloth to 3D architecture

An ordinary fabric is a *sheet*: however complex its weave, its yarns live essentially in one plane, and stacking sheets to make thickness (a laminate of plies) leaves the stack's weakest direction — through-thickness — held together only by resin or adhesive. A **3D fabric** builds reinforcement into the third direction during formation. The family divides by how the through-thickness (Z) connection is made:

- **Multilayer fabrics:** two or more woven layers formed together and **stitched by their own warp or weft** (self-stitched double cloth is the simplest ancestor — TE-3205) or by an extra binder yarn system.
- **Orthogonal 3D weaves:** separate X (weft), Y (warp) and Z (through-thickness) yarn systems: layers of straight warp and weft are bound by Z yarns travelling vertically between the outer layers. The in-plane yarns stay *straight* (no crimp), which is the composite engineer's dream and the weaver's hard problem.
- **Angle-interlock weaves:** warp yarns travel diagonally through the thickness, binding several weft layers on their way — through-thickness interlock (binder passes through all layers) or layer-to-layer interlock (binder links adjacent layers only).
- **3D knitted fabrics:** spacer fabrics (Topic 2 — two faces with a Z-direction pile) and shaped/whole-garment knitting that forms tubes, branches and three-dimensional shells by stitch transfer and partial knitting.
- **3D braids and tri-axial structures** (Topic 1) complete the family.

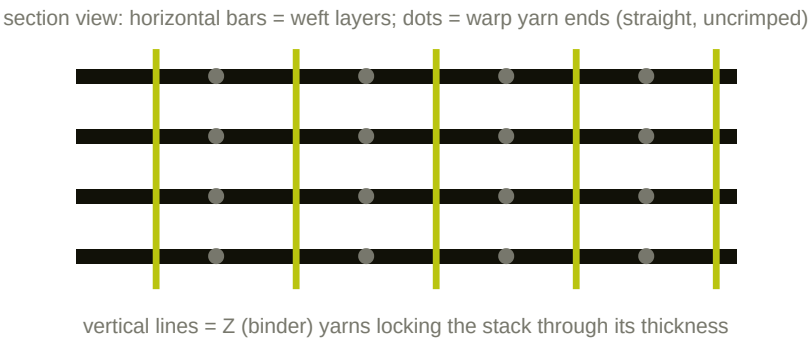


Figure 1 — Orthogonal 3D weave in section (schematic): straight warp layers and straight weft layers form the in-plane reinforcement; separate Z yarns bind the stack, so no ply can peel away without breaking yarn.

2. Production

3D weaving uses modified looms: multiple warp layers are presented as a stack of sheds, weft is inserted layer by layer (or by multiple shuttles/rapiers), and the Z yarn — often the warp of one system re-routed, or a dedicated binder warp on its own beam and shedding — travels through the stack. Conventional looms make multilayer and layer-to-layer interlocks; fully orthogonal structures need purpose-built 3D looms with individual control of many warp layers. Beat-up must compact a stack, not a sheet, and take-up handles a thick, heavy product. **3D knitting** splits between spacer machines (Topic 2) and flat-bed machines whose transfer and short-row (partial knitting) capabilities shape shells and tubes — the fully fashioned logic of FM-4201 extended into three dimensions. In both routes the governing production fact is the same: **shape and thickness are programmed into the machine settings**, so tooling cost sits in design and programming rather than in cutting dies and sewing lines.

WORKED EXAMPLE — MASS AND FIBRE ACCOUNT OF A 3D PREFORM PANEL

An orthogonal preform is built of 5 identical in-plane layers, each contributing 200 g/m² of warp + weft, plus Z-binder yarns contributing 60 g/m².

Total areal mass = 5 × 200 + 60 = **1060 g/m²**; the binder is 60 ÷ 1060 ≈ **5.7%** of the mass.

That small Z fraction is the whole point of the exercise: under 6% of the material, placed exactly where a 2D laminate is weakest, converts a stack that can delaminate into a single structure that cannot.

3. Properties

- **Delamination resistance:** through-thickness yarns must be broken or pulled out for layers to separate — impact damage stays local instead of spreading as ply peeling; the headline property for composite users.

- **Straight in-plane yarns (orthogonal types):** no crimp means the fibres' full modulus and strength are available in-plane — contrast FM-4203's obliquity account for crimped yarns — and resin infusion meets a regular, open architecture.
- **Near-net shape:** T-, I- and box-section profiles, tubes and shaped panels woven or knitted to form, eliminating cutting, joining and their waste and weak points.
- **Energy absorption and damage tolerance:** the Z system and layer architecture absorb impact progressively — crash structures, ballistic and protective panels.
- **Costs of the architecture:** slower production, expensive machines and programming, limited width and design flexibility, and properties that are anisotropic by design and must be engineered per part — 3D is a solution to specific structural problems, not a universal upgrade.

4. End-uses

Composite reinforcement for aerospace and automotive parts (stiffeners, panels, engine and crash structures), ballistic and impact protection, building and geotechnical reinforcement, medical implants and scaffolds (shaped knits and woven tubes), footwear and apparel shells (knitted uppers are mass-market 3D knitting), inflatable and deployable structures, and filtration and spacer duties where the 3D knit of Topic 2 is the economic member of the family.

Glossary

Orthogonal weave — 3D weave with straight X, Y yarns bound by Z yarns at right angles. **Angle interlock** — binder warp travelling diagonally through some or all layers. **Layer-to-layer / through-thickness interlock** — binder linking adjacent layers only / the full stack. **Delamination** — separation of plies in a layered structure. **Near-net shape** — product formed essentially to final geometry. **Preform** — dry textile reinforcement later impregnated with resin.

Key points

• 3D = a yarn system working through the thickness, by multilayer stitching, orthogonal Z-binders, angle interlock, or knitted spacing/shaping.
• A few percent of Z yarn (worked: 5.7%) removes the laminate's weakest direction.
• Straight in-plane yarns keep full fibre properties; shape is a machine programme, not a cutting plan.
• The price is speed, machine cost and design effort — specify 3D where delamination, impact or shape justify it.

Practice

1. An aircraft panel made from stacked 2D plies fails by delamination after impact. Explain the failure and the 3D-structural cure, naming the yarn system responsible.
2. Compute the areal mass and Z-binder percentage of a preform of 4 layers at 220 g/m² each plus 55 g/m² of binder.
3. Distinguish through-thickness from layer-to-layer angle interlock, and state one consequence for damage tolerance.
4. Why do orthogonal preforms give stiffer composites than plain-weave preforms of the same fibre and areal mass? Answer through crimp.

Answer key

1. Impact drives cracks *between* plies, where only resin holds the stack; the cracks spread under further load and the panel loses stiffness without any fibre breaking. In an orthogonal 3D preform the **Z-binder yarns** cross every interface: a delamination crack must break or extract yarns to advance, so damage stays confined near the impact and residual strength is far higher.
2. Total = 4 × 220 + 55 = **935 g/m²**; binder share = 55 ÷ 935 ≈ **5.9%**.
3. In **through-thickness** interlock each binder warp pass crosses the whole stack, tying the outer layers directly together; in **layer-to-layer** interlock a binder links only neighbouring layers, chaining the stack. Consequence: through-thickness binding resists a single delamination front crossing the whole part more directly, while layer-to-layer gives a somewhat more drapeable, less surface-disturbed preform whose layers can still shift stepwise.
4. Plain-weave yarns are crimped: under in-plane load the first response is crimp removal (obliquity, FM-4203), so the effective modulus is low until the yarns straighten, and the fibres never work purely axially. Orthogonal preform yarns are laid **straight**, so load meets fibre modulus immediately — same fibre, same mass, higher stiffness, purely from geometry.
